

← drug candidate is effective in phase 2, it can be moved quickly into phase 3, using the same patients.

Lung-MAP was the result of private organizations, government agencies, research institutions, and the pharmaceutical industry working toward the common goal of improving clinical trials. Among events leading to the study were the 2012 Friends of Cancer Research-Brookings Conference on Clinical Cancer Research as well as the 2012 establishment of an NCI Thoracic Malignancy Steering Committee task force charged with developing a series of master lung cancer protocols.

ALCHEMIST Study

In August 2014, the NCI announced the launch of a second study designed to test specific targeted drugs in patients with lung cancer. This study seeks to learn whether drugs known to prolong survival in patients with advanced-stage lung cancer might also help those with early-stage disease. Known as the Adjuvant Lung Cancer Enrichment Marker Identification and Sequencing Trials, or ALCHEMIST, the study involves 3 integrated trials that will identify patients with early-stage lung cancer with tumors that have uncommon genetic changes. Researchers will then test drug treatments targeted toward those alterations to determine whether they improve survival.



We'll have something to offer every patient with lung cancer. Our goal is to have the right drug for the right patient at the right time. — Roy Herbst, MD, PhD

“ALCHEMIST is one of the first studies being launched in lung cancer or any other cancer in the adjuvant setting as part of NCI's precision medicine initiatives,” says Shakun Malik, MD, head of thoracic cancer therapeutics in the NCI's clinical investigations branch.

The trial, supported by the NCI, will be coordinated by the Alliance for Clinical Trials in Oncology and the Eastern Cooperative Oncology Group-American College of Radiology Imaging Network (ECOG-ACRIN) Cancer Research Group. All of the National Clinical Trials Network groups collaborated on its development and will participate in the studies. “We could not have done this as individual cooperative groups,” says Geoffrey Oxnard, MD, a thoracic oncologist at Dana-Farber

Cancer Institute in Boston, Massachusetts, who is helping lead the screening phase of the trial. “We can only launch this trial because we're now one national network.”

Patients will be recruited at multiple centers during the next 5 to 6 years. To qualify, they must have already undergone surgery to remove their lung tumors as well as completed adjuvant treatment, including chemotherapy and radiation. The first component trial will involve the screening of 6000 to 8000 patients. Tumor samples will be analyzed for alterations in the epidermal growth factor receptor (EGFR) gene (found in approximately 10% of patients with adenocarcinoma of the lung) and the anaplastic lymphoma kinase (ALK) gene (found in 5% of patients with lung adenocarcinoma). Tissue also will be taken from patients with neither of these mutations and analyzed to learn more about how these cancers behave in earlier stages of the disease, Dr. Malik says.

In the other 2 ALCHEMIST trials, patients with EGFR mutations will be treated with erlotinib, whereas patients with the ALK mutation will receive crizotinib. Each trial also will include a placebo arm. The goal is to determine whether these drugs prevent disease recurrence and prolong survival. “There is a growing amount of data showing that these drugs could make a difference in the adjuvant setting, but none of the previous studies has been empowered to ask whether we can change survival in each of these mutated patients,” Dr. Oxnard says.

He adds that many other genetic mutations are believed to be important in the disease and that if researchers can lay the groundwork, additional biomarker studies in the adjuvant setting will occur. “We hope this study can grow and evolve into new arms in this platform once we prove the feasibility of such a big effort,” he says.

ALCHEMIST also will enable researchers to study the characteristics of lung cancer recurrence after adjuvant treatment, which has not previously been examined, Dr. Oxnard says. One concern he and his colleagues have is whether using these drugs in the adjuvant setting could potentially accelerate resistance to treatment. “One of our questions is whether it's better to give these drugs up front or after patients have developed resistance,” he says.

Reference

1. Institute of Medicine. A National Clinical Trials System for the 21st Century: Reinvigorating the NCI Cooperative Group Program. Washington, DC: National Academies Press; 2010.

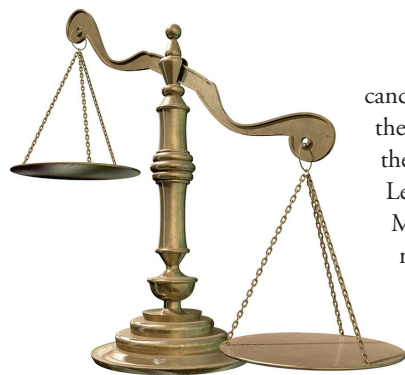
DOI: 10.1002/cncr.29077

Race a Factor in Incidence of Triple Negative Breast Cancer

Regardless of socioeconomic status, black women were nearly twice as likely as white women to be diagnosed with triple negative breast cancer, according to an analysis of a large nationwide dataset.¹ The analysis also found that Asian/Pacific Islander women were more likely to be diagnosed with human epidermal growth factor receptor 2 (HER2)-overexpressing breast cancer.

Some previous studies have suggested that lower socioeconomic status is tied to the higher odds of unfavorable prognoses noted among women in minority racial and ethnic groups; however, these studies were limited by small and incomplete data sets.

The current study analyzed data from 260,174 breast cancer cases in the National Cancer Data Base, a national hospital-based



cancer registry jointly sponsored by the American Cancer Society and the American College of Surgeons. Led by Helmhut Sineshaw, MD, MPH, the research demonstrated that patients of low socioeconomic status had a higher rate of triple-negative breast cancers compared with patients of high or moderate socioeconomic

status. However, when researchers controlled for socioeconomic status, the difference remained. They found that black women were 1.84 times more likely to be diagnosed with triple-negative

breast cancer than white women. They also found that at every level of socioeconomic status, Asian/Pacific Islander women had higher odds of being diagnosed with HER2-overexpressing breast cancer.

Dr. Sineshaw notes that the excess odds of triple-negative breast cancer noted among black women versus white women at every socioeconomic level indicate that further studies are needed to identify which additional factors contribute to these differences.

Reference

1. Sineshaw HM, Gaudet M, Ward EM, et al. Association of race/ethnicity, socioeconomic status, and breast cancer subtypes in the National Cancer Data Base (2010 to 2011). *Breast Cancer Res Treat*. 2014;145:753-763.

DOI: 10.1002/cncr.29078

Obesity Associated with Higher Mortality in Women with ER-Positive Breast Cancer

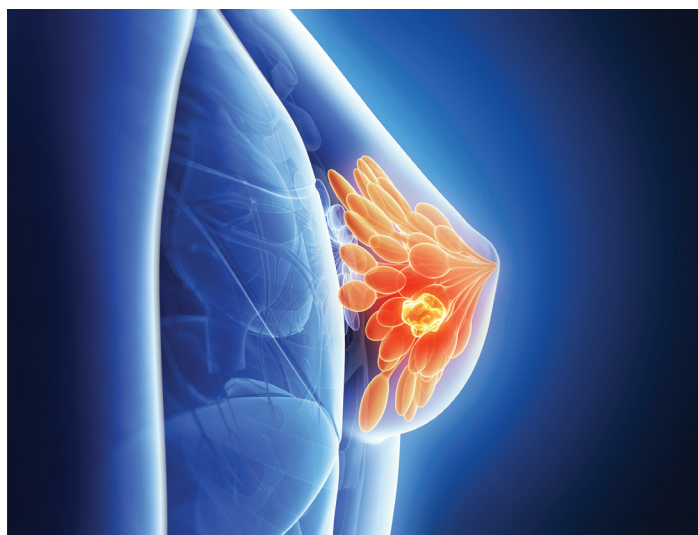
During the oral abstract session on breast cancer at the 2014 annual meeting of the American Society of Clinical Oncology, which was held May 29 to June 3 in Chicago, Illinois, a researcher noted that obesity appears to contribute to negative outcomes in premenopausal and perimenopausal women with estrogen receptor-positive breast cancer.

Hongchao Pan, PhD, of the University of Oxford in the United Kingdom, said that data from 80,000 women in 70 clinical trials analyzed by the Early Breast Cancer Trialists' Collaborative Group demonstrated definite, independent adverse effect on breast cancer mortality among these groups of patients. Conversely, no correlations were found between obesity and long-term survival in postmenopausal women with either ER-positive or ER-negative disease.

Dr. Pan and his colleagues observed that obesity increases the risk that the breast cancer will recur within the next 10 years by approximately one-third. The breast cancer mortality rate was 21.5% after 10 years among obese premenopausal women with ER-positive disease, compared with 16.6% among normal-weight women with the disease. Breast cancer-associated mortality also was found to be increased among women with higher body mass index (BMI) scores.

Obesity is defined as a BMI of 30 kg/mm² or higher. Dr. Pan says researchers were surprised at the results, and had expected to observe the same association in women with ER-positive postmenopausal disease. In that group of 40,000 women, only those with a BMI of 40 kg/m² or more demonstrated a higher likelihood of breast cancer mortality. In the 20,000 postmenopausal women with ER-negative disease, a slightly increased risk of mortality was noted among those who had a BMI of 40 kg/m².

Sharon Giordano, MD, MPH, of The University of Texas MD Anderson Cancer Center in Houston, had several comments on the study, including that increased BMI likely



affects tumor proliferation, increased angiogenesis, and cell survival.

Dr. Giordano mentions a few other issues in the study, including:

- The nonrandomized nature of the trial may have led to an imbalance in the baseline characteristics.
- No adjustment was made for race and/or ethnicity, which may have affected the metabolic activity and outcomes in the participants.
- Because clinical trial participants tend to be younger, better educated, and healthier, the obesity rates in the study population versus those in the general population (23% and 34%, respectively) may reflect a selection bias.

Dr. Giordano adds that future research should examine whether interventions to help patients manage their weight after diagnosis will help affect long-term outcomes.

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